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## Experimental Evaluation of Vibration Attenuation Using a Frequency-Adaptive Damper in a Large Van

--Manuscript Draft--

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# Experimental Evaluation of Vibration Attenuation Using a Frequency-Adaptive Damper in a Large Van

## Abstract

Road induced vibration is a major contributor of vehicle ride quality and occupant vibration exposure, particularly under transient and irregular road excitation. The suspension damper plays a critical role in controlling the transmission of these vibrations, while conventional dampers with fixed damping characteristics may not provide optimum attenuation over a broad range of excitation conditions. This study experimentally evaluates the vibration attenuation performance of a frequency adaptive damper (FAD) installed in a large van and compares it with a conventional passive damping configuration under discrete bump and unpaved road excitation. Vehicle vibration was assessed using frequency weighted root mean square (wRMS) acceleration, vibration dose value (VDV), vibration transmissibility, and power spectral density (PSD). A DEP–CEEMDAN–MPE based decomposition was additionally employed to identify ride frequency vibration components and quantify the energy associated with the resulting intrinsic mode functions (IMFs). Under bump excitation, the FAD reduced the cervix-region wRMS, VDV, and axle to cervix transmissibility by 34.0%, 54.3%, and 33.5%, respectively, demonstrating substantial attenuation of the transient vibration transmitted toward the occupant. Under unpaved road excitation, the FAD maintained consistent vibration attenuation, with reductions of up to 16.7% in wRMS and 20.2% in VDV at the investigated occupant lumbar and cervix spine locations. PSD and CEEMDAN analysis further indicated that the attenuation was concentrated within important low frequency vibration components. The results demonstrate that frequency adaptive damping technology can provide improved vibration isolation under both transient and broadband road excitation and can potentially enhance ride quality while reducing occupant whole-body vibration exposure.

## Keywords

Frequency-adaptive damper; Automotive suspension; Vehicle vibration; Ride comfort; Whole-body vibration; Vibration transmissibility; Power spectral density; CEEMDAN; IMF energy

## 1. Introduction

The human body exhibits frequency dependent biodynamic responses, with increased vibration transmissibility occurring near specific resonant frequencies. Under uneven road conditions the vehicle and its occupants are subjected to increased road excitations and the resulting higher vibration transmission results in reduced comfort and there by harsh ride. Prolonged hard ride and exposure to severe vibrations adversely affect the human health and its severity depends on vibration magnitude, frequency and duration [1]. The long-term exposure of such vibration cause spine damage resulting from the changes the phycological structure. [2]. For seated occupants, whole-body vibration exhibits a pronounced frequency dependence, with maximum vibration responses reported in the 3–6 Hz range. Segmental transmissibility analyses further indicate elevated vibration transmission to major body segments, particularly the pelvis, within approximately 4–6 Hz [3]. ISO 2631 [4] considers whole-body vibration within the 1–80 Hz frequency range for the assessment of its effects on human health, comfort, and vibration perception. Vehicle ride comfort using the chassis acceleration is evaluated in the frequency range up to 20Hz [5]. Out of these frequencies 5 to 20 Hz might cause spine damage [6], a fall in blood pressure by 8–10 Hz range, possibility of peripheral vasoconstriction by 11–12 Hz, and frequencies over 12 Hz can cause muscle twitching, induced arrhythmia, increase in pain and bleeding [7].

The implications of vibrations and its long-term exposure are profoundly important in medical emergency vehicles, where occupants will be critically injured, ill or neonatal. Studies of medical emergency transport has shown that excessive vibration transmitted though chassis to stretcher can create resonant modes in human body organs amplifying the associated health hazards [8]. Several studies indicate clear association of neonatal health hazards and injuries relating to the vibration exposure during interfacility transport [9, 10].

The suspension system comprising spring and dampers isolate the chassis from road induced vibrations while providing vehicle stability through road holding and minimizing the body roll. Air springs can increase ride comfort due to their ability to bring down natural frequencies below 5Hz [11]. However, the spring primarily determines the suspension stiffness and natural frequency, whereas the damper governs the dissipation of vibratory energy and strongly influences the amplitude and transient response of the vehicle body. In essence, it is the damper that determines the rate of oscillation decay, and how a vehicle reacts to the road profile. The hydraulic damper arrests these vibrations by converting the kinetic energy of motion into heat

by forcing the damping fluid across orifices [12]. Conventional dampers with fixed damping characteristics may not provide optimum vibration attenuation over the wide range of excitation frequencies encountered during vehicle operation [13]. Advanced damping systems address this limitation by modifying the damping force through changes in hydraulic flow and valve characteristics. Active and semi-active dampers can employ electronically controlled mechanisms to vary the damping characteristics according to the road excitations, whereas passive position-sensitive, stroke-sensitive, and frequency-dependent valving systems achieve motion dependent damping through mechanically designed hydraulic flow paths and valve mechanisms. Among these approaches, frequency-based damping is particularly attractive due to its better control over lower frequencies while retaining a predominantly passive hydraulic architecture [14].

Vehicle vibration is commonly evaluated using a combination of time and frequency domain measures. Root mean square (RMS) acceleration provides a measure of the overall vibration magnitude, while frequency weighted RMS acceleration accounts for the frequency dependence of human sensitivity to whole-body vibration [4]. Power spectral density (PSD) analysis characterizes the distribution of vibration energy over frequency and identifies the dominant frequency regions associated with vehicle and suspension dynamics [15]. For vibration exposures containing transient or intermittent events, the vibration dose value (VDV) provides a cumulative measure that is less sensitive to the limitations of RMS alone [16]. In suspension assessment, vibration transmissibility is additionally used to quantify the transmission of vibration between the excitation and response locations [17]. Although these measures provide complementary information on vibration magnitude, frequency content, cumulative exposure, and transmission, each represents the vibration response through a different projection of the measured signal. In particular, PSD and RMS-based measures do not explicitly describe the temporal evolution of individual frequency components in non-stationary vibration, while VDV provides a cumulative exposure measure without identifying the frequency-specific origin of the contribution. This motivates the use of adaptive frequency resolved decomposition analysis alongside conventional vibration metrics.

Sun et al. [18] used a decomposition-based analysis framework, DEP–CEEMDAN–MPE–INHT, for the analysis of noisy blasting seismic waves. In the present study, we propose a DEP–CEEMDAN–MPE-based IMF energy framework for the comparative evaluation of automotive suspension vibration performance. The framework incorporates dual endpoint processing (DEP) to mitigate endpoint effects, complete ensemble empirical mode

decomposition with adaptive noise (CEEMDAN) to decompose complex vibration signals into intrinsic mode functions (IMFs), and multiscale permutation entropy (MPE) to characterize the decomposed vibration components. The resulting IMFs are quantified using their integrated squared acceleration to determine the vibration contribution associated with individual frequency components, thereby providing a frequency-resolved comparison of the frequency-adaptive damper and conventional passive suspension. The proposed DEP–CEEMDAN–MPE–IMF energy framework is employed as a complementary diagnostic approach alongside established vibration metrics, including PSD, RMS, frequency-weighted RMS, VDV, and transmissibility, to identify the frequency regions associated with changes in vibration transmission.

Large vans are also widely used as platforms for special-purpose and emergency vehicles, where the vibration transmitted to the vehicle structure and occupants can have important consequences for occupants' health and comfort. Recent investigations of ambulance transport have demonstrated that road-induced vibration is transmitted through the vehicle floor and patient support system, with several system modes occurring within frequency ranges relevant to human body dynamics. Moreover, increasing the damping of the support system has been shown to reduce vibration propagation from the vehicle to the occupant support system, highlighting the importance of suspension and damping characteristics in controlling transmitted vibration [19]. However, recent automotive research on advanced damping systems has largely focused on damper-level characterization, component development, and numerical or reduced-order vehicle assessments, with comparatively limited experimental investigation of their effects on vibration transmission at multiple vehicle and occupant-related locations under realistic road excitation. Therefore, there is a need for experimental vehicle level whole body vibration (WBV) investigations to evaluate how advanced damping systems influence vibration transmission to the vehicle structure and occupants under realistic operating conditions. In the present study, this need is addressed through an experimental evaluation of frequency-adaptive damper (FAD) against a conventional passive damping configuration in a large van.

The primary objective is to compare the vibration attenuation of the two suspensions using complementary vibration assessment metrics, including power spectral density (PSD), root-mean-square (RMS) acceleration, frequency-weighted RMS acceleration, vibration dose value (VDV), and vibration transmissibility. To further resolve the frequency-dependent and non-stationary characteristics of the measured vibration, a DEP–CEEMDAN–MPE-based IMF

energy analysis is employed as an additional frequency resolved decomposition layer. The analysis considers vibration responses at multiple locations within the vehicle and occupant environment, thereby enabling the attenuation provided by the FAD to be evaluated not only in terms of overall vibration severity but also in terms of the frequency resolved vibration characteristics. The study therefore provides a multi-dimensional experimental framework for relating damper characteristics to vibration attenuation across vehicle structural and occupant-related locations, while retaining standardized whole-body vibration (WBV) metrics as the primary measures of ride and exposure severity.

## 2. Vibration Evaluation Metrics

All occupants in the road vehicle expose themselves to WBV and the human response to these WBV depends on frequency, magnitude, direction, and duration of the vibration signal. ISO 2631-1 and BS 6841 provide the general framework to evaluate the human response based on the above-mentioned parameters of the signal. In the present study ISO2631-1 is used. ISO 2631-1 uses frequency-weighted RMS acceleration as a principal measure for evaluating whole-body vibration. The weighted RMS is calculated by applying the ISO based weighting function to the measured acceleration signal and is expressed as,

$$a_{w,i} = \left( \frac{1}{T} \int_0^T a_{\omega,i}^2(t) dt \right)^{1/2} \quad (1)$$

where  $a_{w,i}(t)$  is the frequency-weighted acceleration in the  $i$ -th axis,  $T$  is the measurement duration, and  $a_i(t)$  is the measured acceleration time history before frequency weighting. The cumulative severity of the vibration during transient events is calculated using the vibration dose value (VDV). For the frequency-weighted acceleration  $a_{w,i}(t)$ , the VDV is defined as:

$$VDV = \left( \int_0^T a_{\omega,i}^4(t) dt \right)^{1/4} \quad (2)$$

Unlike RMS, the fourth-power dependence in the VDV expression gives greater influence to high-amplitude transient events. VDV was therefore included to complement the weighted RMS measure for the non-stationary and off-road vibration conditions investigated in this study. Power spectral density (PSD) analysis was used to characterize the distribution of vibration over frequency and to identify dominant frequency regions associated with the vehicle and suspension response. The PSD was calculated from the acceleration time histories

and used to compare the frequency content of the FAD and passive configurations. In contrast to RMS and VDV, which provide scalar measures of vibration severity, PSD identifies the frequency regions contributing to the measured response. PSD is a common methodology used in automotive sector to compare the suspension performance [20,21]. The commonly used technique is Welch's method [22,23]. The vibration transmissibility is another parameter used to quantify the transmission of vibration between selected input and response locations. The frequency-domain transmissibility was expressed as:

$$T(f) = \frac{|A_{response}(f)|}{|A_{input}(f)|} \quad (3)$$

A transmissibility magnitude below unity indicates attenuation between the corresponding locations, whereas a value above unity indicates amplification. This measure was used to assess the effectiveness of the FAD in limiting the propagation of road-induced vibration through the vehicle structure.

Although the above metrics provide complementary information on vibration magnitude, exposure severity, frequency distribution, and transmission, they do not directly resolve the temporal evolution of individual frequency components in non-stationary vibration. To obtain this additional information, a DEP–CEEMDAN–MPE frequency resolved decomposition analysis was performed alongside the conventional measures. This framework combines endpoint processing, CEEMDAN decomposition, multiscale permutation entropy characterization for noisy vehicle vibration signals. In the present study, DEP was applied to mitigate endpoint effects, CEEMDAN was used to decompose the measured acceleration into intrinsic mode functions, MPE was employed to characterize the complexity of the decomposed modes and to assist in identifying meaningful vibration components. The integrated squared acceleration of the selected IMFs was then calculated to quantify their vibration-energy contribution.

Let the measured WBV acceleration be:

$$x(t), \quad 0 \leq t \leq T \quad (4)$$

For discrete measurement:

$$x(n), \quad n = 1, 2, 3, \dots, N \quad (5)$$

The empirical mode decomposition is sensitive to boundary conditions; therefore, the measured signal was extended at both endpoints before decomposition. The endpoint-processing operation can be represented generally as:

$$x_{DEP}(t) = D\{x(t)\} \quad (6)$$

where  $D\{\cdot\}$  denotes the data-extension procedure used to generate physically consistent left and right extensions. Thus  $x(t) \rightarrow x_{DEP}(t)$  with  $N_{DEP} > N$ . The purpose of DEP is to reduce endpoint induced distortion during subsequent CEEMDAN decomposition. The DEP-processed signal is decomposed into intrinsic mode functions:

$$x_{DEP}(t) = \sum_{k=1}^K c_k(t) + r_K(t) \quad (7)$$

where  $c_k(t)$  is the  $k$ th IMF,  $K$  is the number of extracted IMFs,  $r_K(t)$  is the final residual. CEEMDAN uses an ensemble of noise assisted decompositions to obtain the modes. The first IMF is obtained from the ensemble average:

$$c_1(t) = \frac{1}{M} \sum_{j=1}^M IMF_1(t) [x_{DEP}(t) + \epsilon \omega_j(t)] \quad (8)$$

where  $M$  is the number of noise realizations,  $\omega_j(t)$  is the  $j$ th noise realization, and  $\epsilon$  controls the added noise amplitude. The first residual is then:

$$r_1(t) = x_{DEP}(t) - c_1(t) \quad (9)$$

and subsequent IMFs are extracted recursively until the stopping criterion is reached:

$$x_{DEP}(t) = \sum_{k=1}^K c_k(t) + r_K(t) \quad (10)$$

For each IMF  $c_k(t)$ , MPE is calculated to quantify its complexity over multiple temporal scales. At scale  $\tau$ , a coarse-grained sequence can be represented as:

$$y_j^{(\tau)} = \frac{1}{\tau} \sum_{i=(j-1)\tau+1}^K c_k[i] \quad (11)$$

An embedded vector of dimension  $m$  is then constructed:

$$Y_j = [y_j, y_{j+\tau}, \dots, y_{j+(m-1)\tau}] \quad (12)$$

The ordinal patterns of these vectors give probabilities  $p_\pi$ , from which permutation entropy is calculated:

$$PE_k(m, \tau) = - \sum_{\pi} p_{\pi} \ln p_{\pi} \quad (13)$$

The normalized value is

$$MPE_k(m, \tau) = \frac{- \sum_{\pi} p_{\pi} \ln p_{\pi}}{\ln(m!)}, \quad 0 \leq MPE_k \leq 1 \quad (14)$$

For each IMF, the MPE value was evaluated over multiple scales to characterize the complexity of the decomposed component. The MPE was used as a supporting criterion for distinguishing structured vibration components from highly irregular components, while the final IMF selection was based on the combined consideration of MPE, dominant frequency, and vibration-energy contribution. This approach avoids relying solely on IMF order or on a fixed entropy threshold, which may not be appropriate for deterministic automotive vibration components. For each selected IMF, the energy is calculated as:

$$E_k = \int_0^T c_k^2(t) dt \quad (15)$$

In discrete form:

$$E_k = \frac{1}{F_s} \sum_{n=1}^N c_k^2[n] \quad (16)$$

where  $F_s$  is the sampling frequency. Since  $c_k$  is in  $\text{m/s}^2$ , the  $E_k$  will be in  $\text{m}^2/\text{s}^3$ .

The combination of these conventional vibration metrics and the decomposition-based analysis enable the performance of the FAD to be evaluated from complementary perspectives, including overall vibration severity, cumulative vibration exposure, vibration transmission, frequency distribution, and frequency-resolved vibration content. The objective of the present study is to determine whether the frequency-adaptive damping characteristics provide measurable attenuation under both transient and broadband road excitation, and to identify the

frequency regions in which such attenuation is most pronounced. To achieve this, controlled bump-road and unpaved-road tests were conducted on a large van, with vibration measurements obtained at the chassis and occupant-related locations. The experimental test configuration and measurement procedure are described in the following section.

### 3. Test Setup

#### 3.1 Test Vehicle

A Force Traveller 3350, a large van commonly used for passenger and emergency medical transportation in India, was selected as the test vehicle. The vehicle is equipped with a conventional leaf-spring suspension at the front axle and a rear electronically controlled air-suspension system (R-ECAS) incorporating twin air springs developed by Rambal Pvt Ltd. The vehicle is fitted with four hydraulic dampers. For the experimental comparison, the conventional hydraulic dampers were replaced by the frequency-adaptive dampers (FADs), and vice versa, while the front leaf springs and rear R-ECAS configuration were kept unchanged. This ensured that the primary difference between the two test configurations was the damping characteristic. The test vehicle is shown in **Figure 1** Force Tempo Traveller test vehicleFigure 1, and the principal suspension parameters are provided in Table 1.



**Figure 1** Force Tempo Traveller test vehicle

**Table 1** Test vehicle suspension parameters

| Parameter                   | Value                  |
|-----------------------------|------------------------|
| Sprung mass                 | 3080 kg                |
| Unsprung mass – front       | 186 kg                 |
| Unsprung mass – rear        | 244 kg                 |
| Spring rate – front         | 54,000 N/m             |
| Spring rate – rear          | 42,550 N/m             |
| Pitch inertia               | 4116 kg m <sup>2</sup> |
| Roll inertia                | 1029 kg m <sup>2</sup> |
| Front-axle distance from CG | 1.571 m                |
| Rear-axle distance from CG  | 1.779 m                |
| Track width – front         | 1.625 m                |
| Track width – rear          | 1.595 m                |

### 3.2 Instrumentation and Damper Configuration

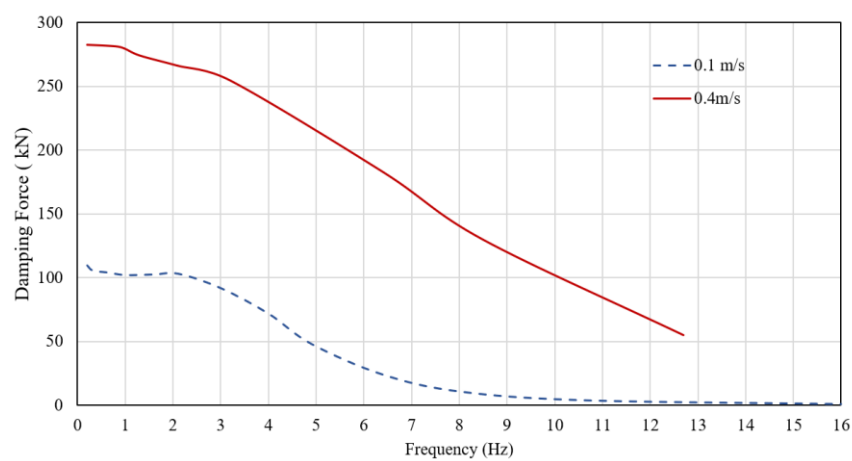
The present study investigates two damping configurations: the OEM conventional passive damper and frequency adaptive damper (FAD) (RNMS-FAD-001) developed and manufactured by Rambal Private Limited, India. The two damper configurations are shown in Figure 2. The FAD constitutes a mechanical valve that modifies the flow path depending on the excitation frequency. The rear R-ECAS and all other suspension components were kept unchanged between the two configurations. Figure 3 shows the damping-force response as a function of excitation frequency at piston velocities of 0.1 and 0.4 m/s, demonstrating the reduction in damper force with increasing excitation frequency characteristic of the FAD. Figure 4 compares the force–velocity characteristics of the FAD and OEM passive damper.

Vehicle vibration was measured using seven ADXL335 accelerometers and a BNO055 inertial measurement unit (IMU). Two accelerometers were mounted at the rear axle, two on the chassis floor, and three at occupant-related locations: the seat, lumbar region, and cervix region. The axle measurements characterize the unsprung mass vibration and the chassis, lumbar and cervix measurements characterize the sprung mass and occupant related transmitted vibration. The

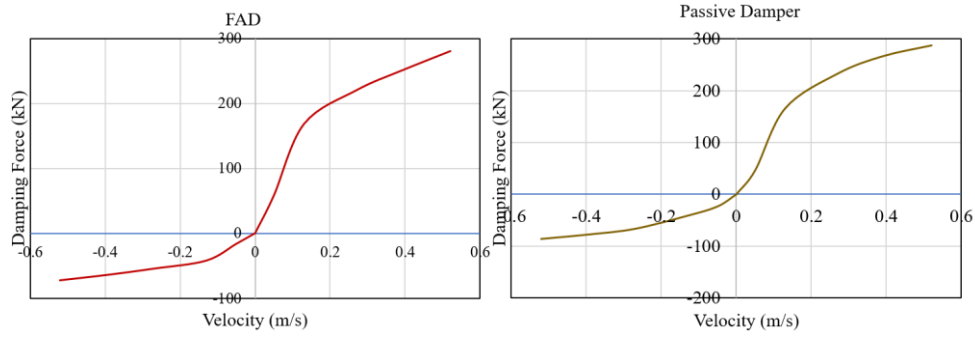
BNO055 IMU was mounted near the vehicle center of gravity to measure vehicle pitch and roll motion. Data were acquired using an Arduino Due at a sampling frequency of 1000 Hz. Sensor locations and the measurement setup are illustrated in Figures 5 and 6.



**Figure 2** Suspension dampers: (a) frequency adaptive damper (FAD); (b) conventional passive damper (OEM)



**Figure 3** FAD damping force vs excitation frequency at fixed piston velocities



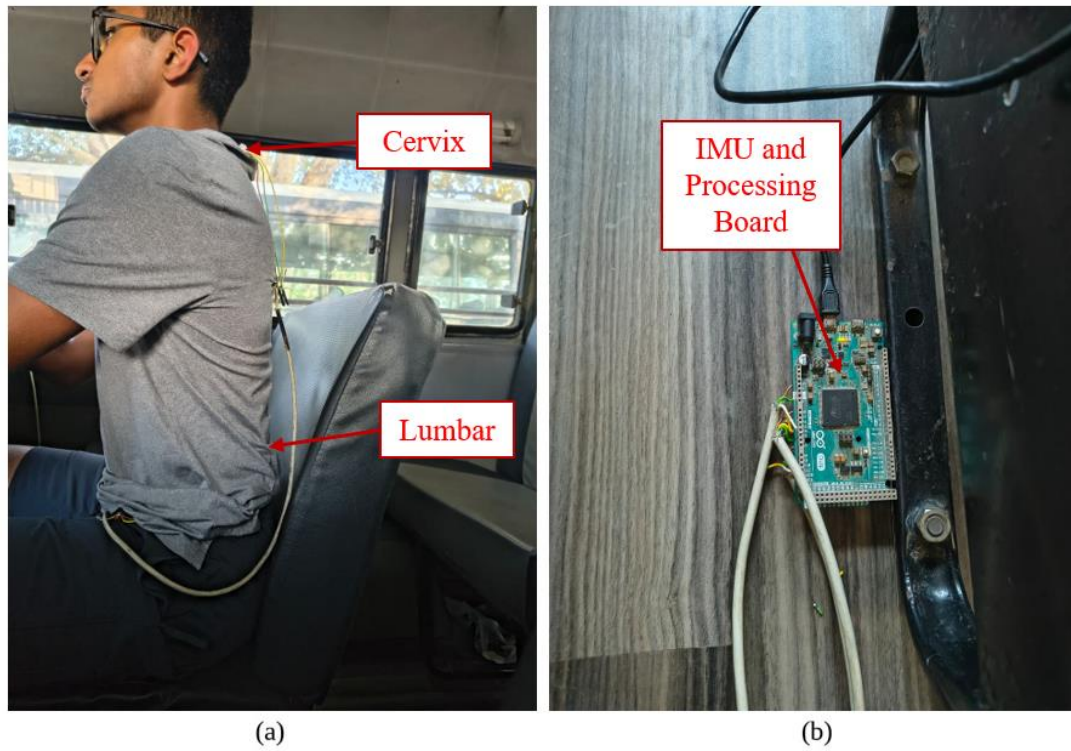
**Figure 4** Comparison of force-velocity characteristics of FAD and passive damper

### 3.3 Instrumentation and Data Acquisition

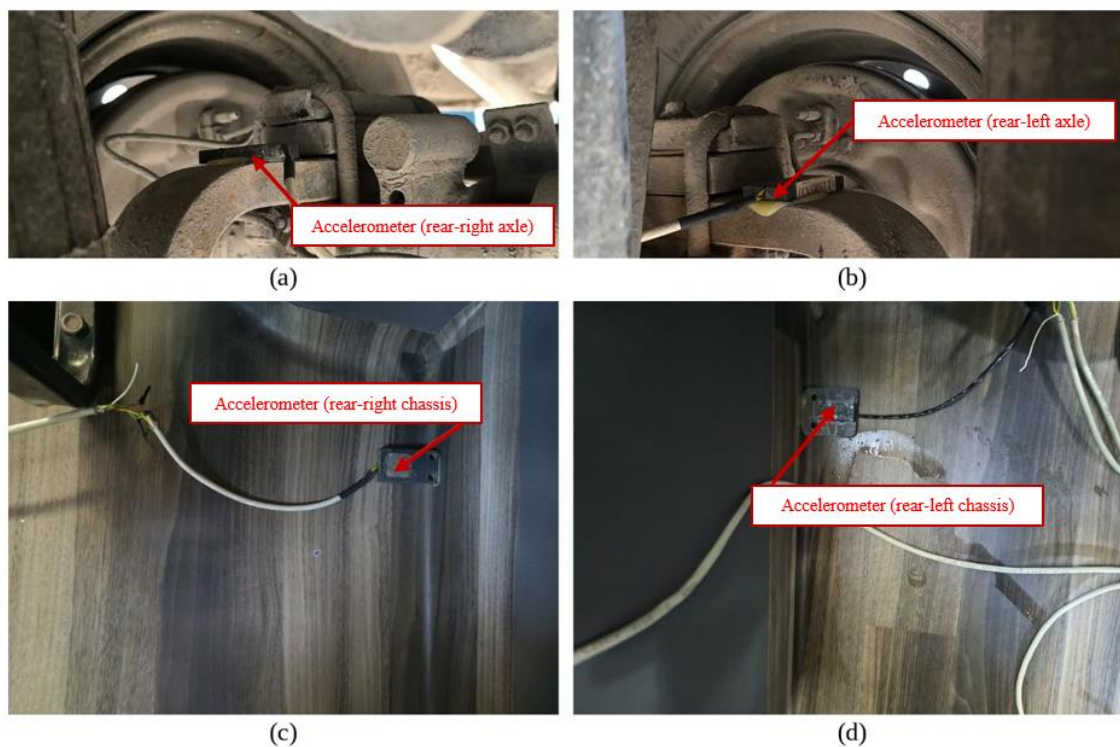
Vibration and vibration transmission from the road to the seated occupant is measured using an array of ADXL335 accelerometers and a single inertial measurement unit (IMU). Two accelerometers are mounted at the rear axle (left and right) to capture the unsprung mass motion, and two accelerometers are mounted on the chassis floor. Three more accelerometers are placed on the occupant at different points: the seat, lumbar spine, and neck. This allows for the vibration transmission to be tracked as it transfers from the road to through the seated body rather than at a single point. The inertial measurement unit BNO055 is mounted approximately near the vehicle's center of gravity to record pitch and roll motion. The sensor specifications are given in Table 2. Data were acquired using an Arduino Due, with a sampling frequency of 1000Hz. Images of the test occupant and Arduino Due board are given in Figure 5. Figure 6 contains labelled images of the accelerometers in the RR and RL chassis and axle positions.

**Table 2** Specifications of the ADXL335 accelerometer and BNO055 IMU

| Sensor  | Type                        | Range                         | Sensitivity / Resolution               | Sampling Frequency |
|---------|-----------------------------|-------------------------------|----------------------------------------|--------------------|
| ADXL335 | 3-axis analog accelerometer | $\pm 3$ g                     | 300 mV/g                               | 1000 Hz            |
| BNO055  | IMU                         | $\pm 4$ g /<br>$\pm 2000$ °/s | 14-bit accelerometer; 16-bit gyroscope | 100 Hz             |



**Figure 5** (a) Instrumented test occupant showing accelerometer placement at the seat, lumbar, and neck locations; (b) IMU located at the back side of Arduino due mounted near vehicle center of gravity

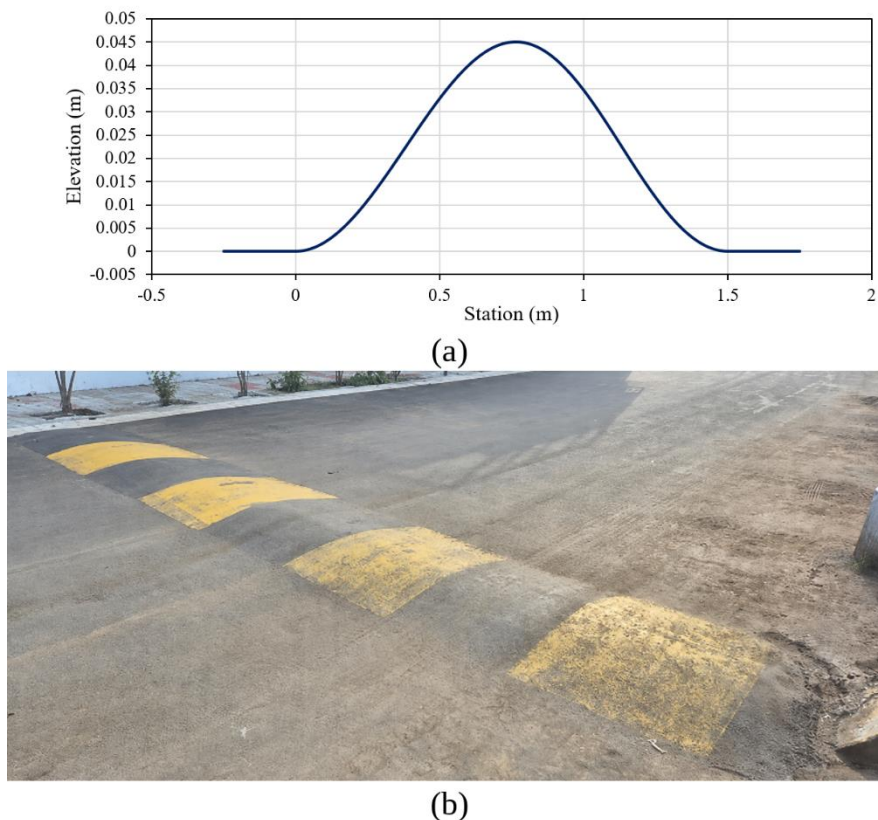


**Figure 6** Accelerometer mounting (a) rear-right axle, (b) rear-left axle, (c) rear-right chassis, (d) rear-left chassis

### 3.4 Road Conditions and Test Procedure

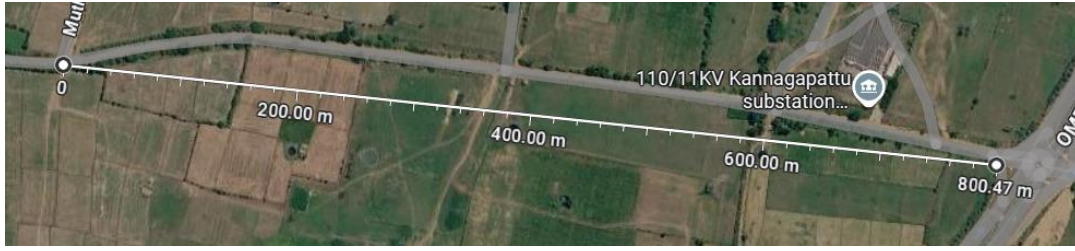
Two road conditions were used to represent possible surfaces that could be encountered by the Tempo Traveller during patient transport, especially in rural areas: a paved road with a bump and an unpaved off-road patch. Both the damper configurations - stock OEM passive damper and FAD were tested in both road conditions, with identical driving procedures followed.

The bump tests were conducted at Rambal Pvt. Ltd.'s suspension test facility. The facility consists of a smooth, paved road with a single bump. The bump is similar in profile to a speed breaker, with a maximum height of approximately 0.045 m and incremental elevation on each side, spanning approximately 1.5m in length. The Tempo Traveller was driven over the bump at a steady speed of 10 km/h, with throttle applied as needed to maintain the speed. The bump geometry is shown in Figure 7.



**Figure 7** (a) measured bump elevation profile; (b) photograph of the bump

The unpaved road tests were conducted over an 800m long unpaved patch of dirt, gravel and rutted surface, which contained rough gradients and a mix of wet and dry sections. The vehicle speed was maintained at 10 km/h. The test section is shown in Figure 8.



**Figure 8** Satellite image of unpaved road test section

Thirty bump-road runs were conducted for each damper configuration, and twenty runs were conducted for each configuration on the unpaved-road section. A representative run was selected from the repeated measurements based on consistent vehicle speed and absence of anomalous disturbances. The same vehicle, suspension configuration, instrumentation, target speed, and test procedure were maintained for both damper configurations to ensure a controlled comparison of their vibration attenuation performance. The acquired acceleration and vehicle motion data were subsequently processed using the evaluation metrics and DEP–CEEMDAN–MPE-based IMF energy framework described in Section 2. The results obtained under bump-road and unpaved-road excitation are presented and discussed in the following section.

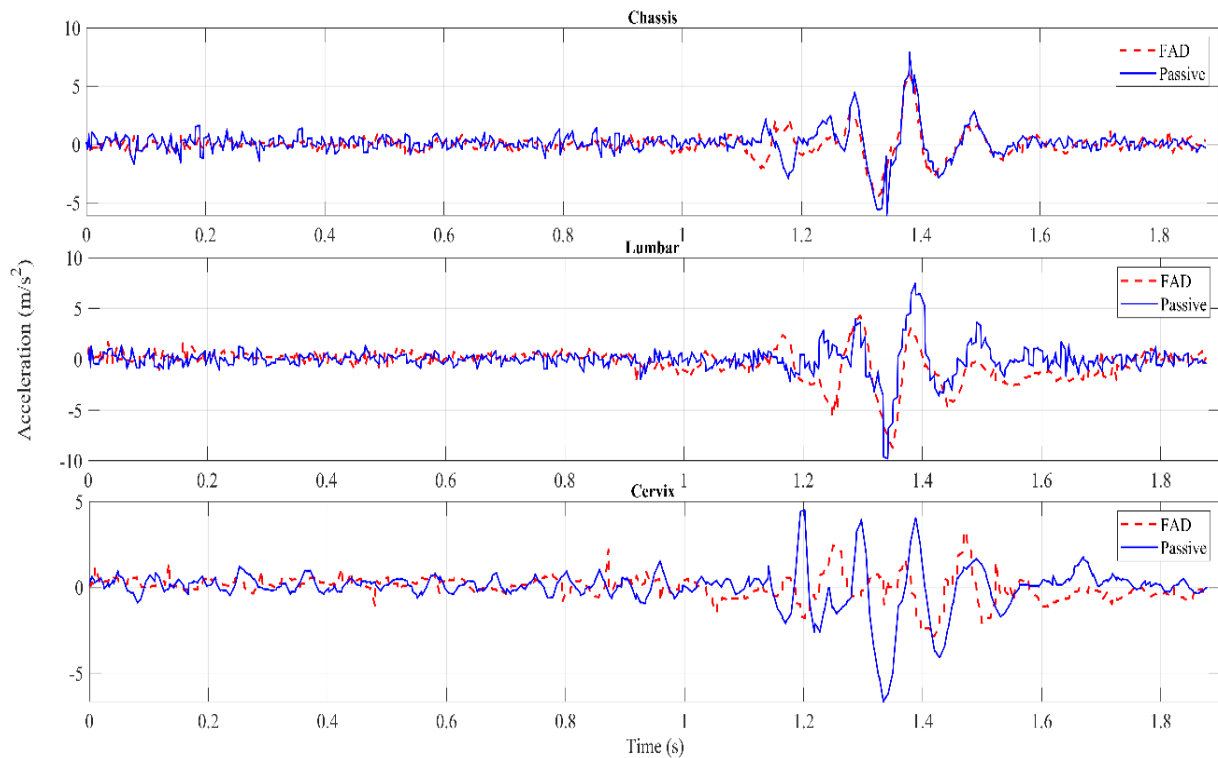
## 4. Results and Discussion

The experimental results are presented in two road excitations: a bump road and unpaved road (off-road condition). These tests were selected to represent two distinct forms of road excitation and WBV for large van. The bump road test represents controlled transient input for evaluating the suspension's ability to attenuate high amplitude excitation and resulting vehicle body oscillations, whereas unpaved road creates off road conditions create irregular and broadband road excitation and is favorable to evaluate wide range of frequency-based conditions and human body response to it. The performance of the frequency adaptive damper (FAD) is compared with the conventional passive configuration using frequency weighted RMS acceleration, VDV, vibration transmissibility, PSD, and DEP–CEEMDAN–MPE based IMF energy analysis, as appropriate for each test condition.

### 4.1. Bump Road Excitation

The bump test results at 10km/h shows the transient response of the suspension systems towards the vibration. Figure 9 represents the measured acceleration responses at the chassis, lumbar, and cervix locations for the passive and frequency-adaptive damper (FAD)

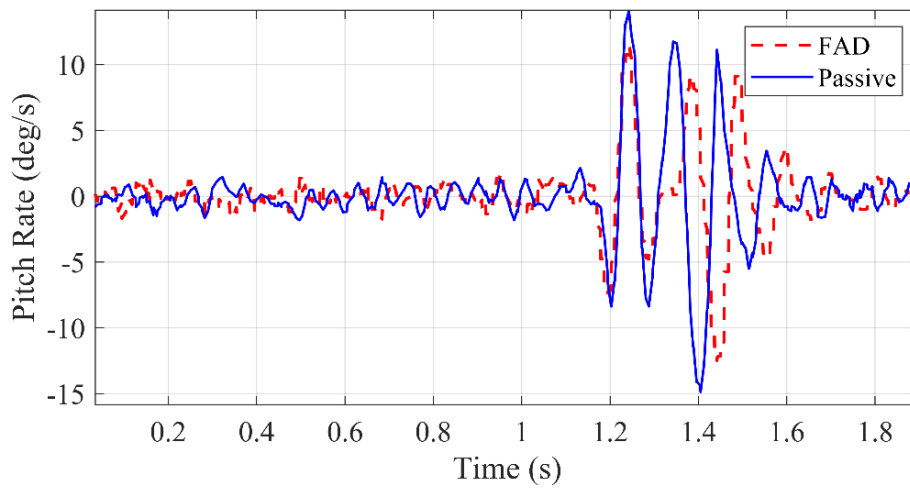
configurations. The responses show a pronounced transient following the bump input, with the magnitude and subsequent oscillatory response differing between the two damping configurations. The FAD response exhibits lower acceleration amplitudes following the excitation, particularly at the chassis and cervix locations, indicating improved attenuation of the transient vibration transmitted through the suspension. The quantitative comparison of the frequency weighted RMS acceleration and VDV further confirms the attenuation observed in the acceleration histories as shown in Table 3 . For the chassis, the frequency-weighted RMS acceleration decreased from 0.624 m/s<sup>2</sup> for the passive configuration to 0.518 m/s<sup>2</sup> with the FAD, corresponding to a reduction of approximately 17.0%. At the lumbar location, the corresponding values decreased from 1.105 to 0.985 m/s<sup>2</sup>, giving a reduction of approximately 10.9%. The most pronounced reduction was observed at the cervix, where the weighted RMS decreased from 0.656 to 0.433 m/s<sup>2</sup>, corresponding to approximately 34.0% attenuation. These results indicate that the benefit of the FAD is location dependent, with the largest reduction occurring at the upper body measurement location.



**Figure 9** Responses subjected to a bumpy road at 10 km/h: Chassis, Lumbar and Cervix regions

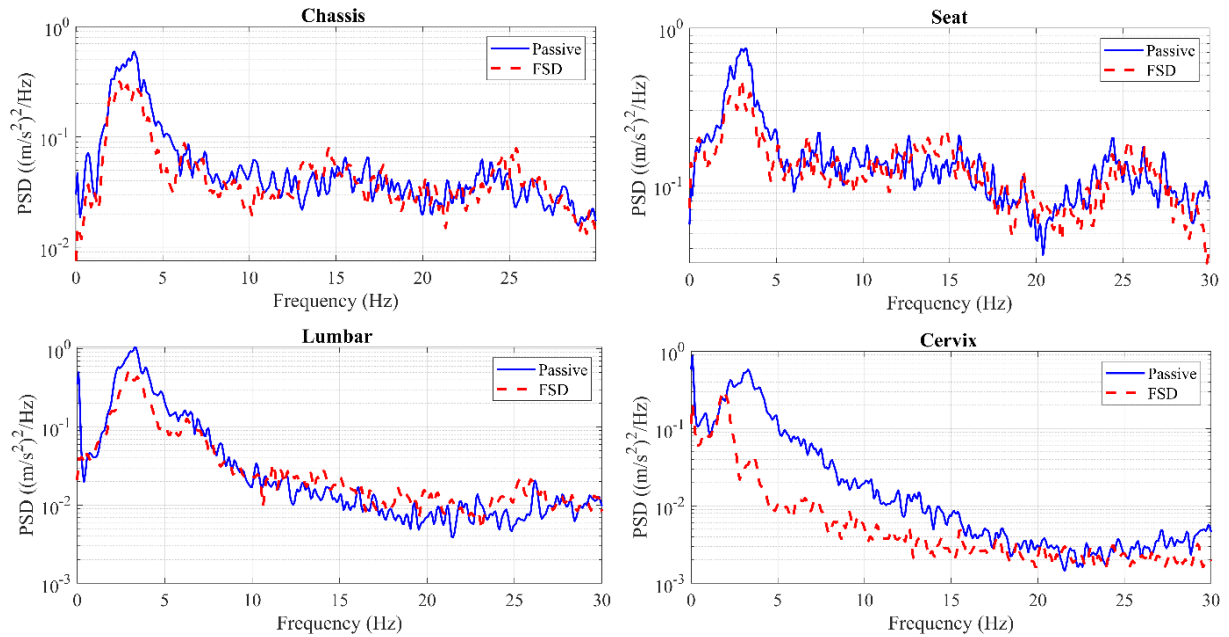
**Table 3** ISO2631-1 based performance comparison of Passive and FAD during bump road test

|         | wRMS    |       |             | VDV     |       |             |
|---------|---------|-------|-------------|---------|-------|-------------|
|         | Passive | FAD   | % Reduction | Passive | FAD   | % Reduction |
| Chassis | 0.624   | 0.518 | 17.0%       | 3.349   | 2.562 | 23.5%       |
| Lumbar  | 1.105   | 0.985 | 10.9%       | 3.956   | 3.537 | 10.6%       |
| Cervix  | 0.656   | 0.433 | 34.0%       | 3.17    | 1.45  | 54.3%       |

**Figure 10** Pitch response for the bump road test

The VDV results provide additional evidence of the effect of the FAD on the transient response. The chassis VDV decreased from 3.349 to 2.562  $\text{m/s}^{1.75}$ , corresponding to a reduction of approximately 23.5%. At the lumbar location, the VDV decreased from 3.956 to 3.537  $\text{m/s}^{1.75}$ , representing a reduction of approximately 10.6%. A substantially larger improvement was obtained at the cervix, where the VDV decreased from 3.170 to 1.450  $\text{m/s}^{1.75}$ , corresponding to approximately 54.3% reduction. The greater reduction in VDV compared with weighted RMS at the cervix indicates that the FAD was particularly effective in reducing the high-amplitude transient components contributing to the cumulative vibration dose at this location. The RMS weighted transmissibility results provide a direct measure of the change in vibration transmission from the axle to the measured response locations. For the axle-to-chassis transmission, the transmissibility decreased from 1.0171 for the passive configuration to 0.8503 with the FAD, corresponding to an attenuation of approximately 16.4%. The axle-to-

lumbar transmissibility decreased from 1.800 to 1.617, corresponding to approximately 10.2% reduction. The largest reduction was again observed for the axle-to-cervix transmission, which decreased from 1.069 to 0.7109, corresponding to approximately 33.5% reduction.



**Figure 11** PSD response of unpaved road test

The bump-road results overall demonstrate that the FAD does not merely reduce the absolute acceleration measured at one location; it reduces the transmission of the transient excitation through the vehicle system. The reduction in pitch response shown in Figure 10 was accompanied by reductions in frequency-weighted RMS acceleration and VDV at the chassis, lumbar, and cervix 1 measurement locations, indicating that the improved vehicle-body response was also reflected in reduced vibration transmission to the occupant-related locations.

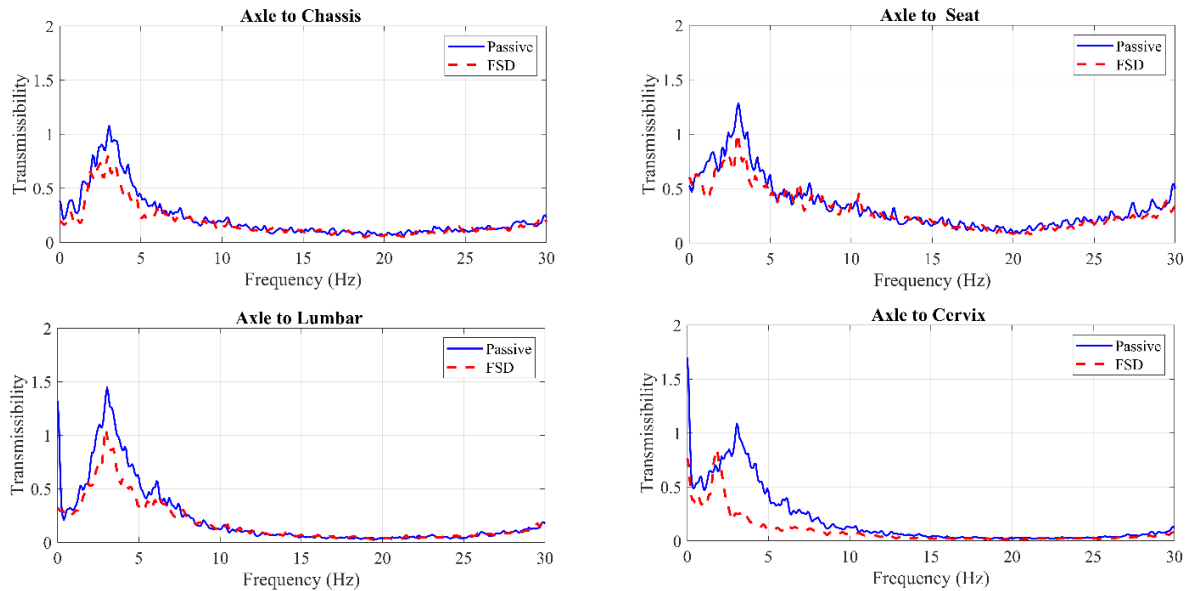
**Table 4** ISO2631-1 based performance comparison of Passive and FAD during unpaved road test

|         | wRMS    |       |             | VDV     |       |             |
|---------|---------|-------|-------------|---------|-------|-------------|
|         | Passive | FAD   | % Reduction | Passive | FAD   | % Reduction |
| Chassis | 1.23    | 1.03  | 16.3%       | 4.58    | 3.771 | 17.7%       |
| Lumbar  | 1.425   | 1.187 | 16.7%       | 5.42    | 4.60  | 15.1%       |
| Cervix  | 1.159   | 0.994 | 14.2%       | 4.245   | 3.386 | 20.2%       |

## 4.2. Unpaved Road Excitation

The unpaved road test response was examined primarily in the frequency domain using PSD, frequency dependent transmissibility, and CEEMDAN based decomposition, followed by frequency weighted RMS and VDV evaluation.

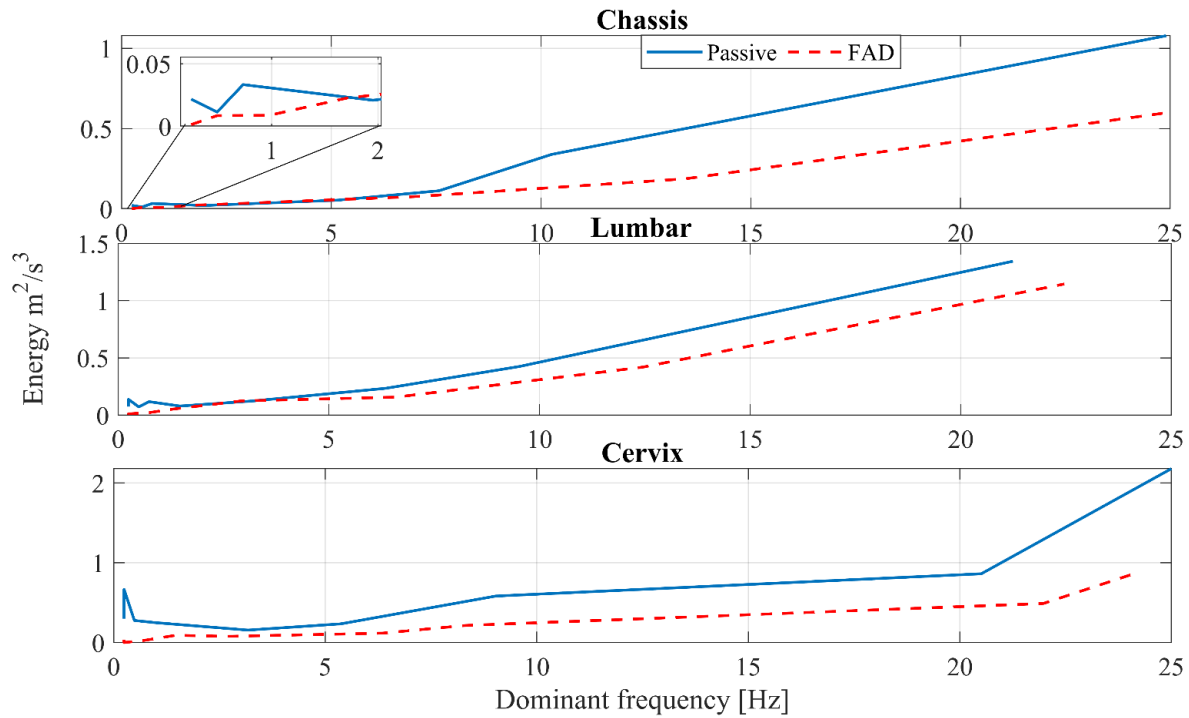
The acceleration PSD of the chassis, seat, lumbar, and cervix measurements for the passive and FAD configurations are shown in Figure 11. It is evident from the figure the FAD outperformed the passive suspension across all regions. Good reduction is observed in the lower 0 to 5Hz region for chassis and seat and better reductions around 4 to 8Hz in lumbar and significant reduction from 3 to 20Hz in cervix. The Table 4 results show a similar trend to the PSD. The frequency weighted RMS results demonstrate a consistent reduction in overall human-relevant vibration magnitude. At the chassis, the weighted RMS showed a 16.3% reduction and a similar trend with 17% reduction in VDV. A consistent trend is followed for lumbar and Cervix with weighted RMS showing 16.7 decrease for FAD I and VDV a 15.1% for lumbar region. The cervix also showed better vibration reduction for FAD compared to passive damper. The PSD, weighted frequency and VDV clearly indicates how FAD attenuates harmful vibrational frequency that is relevant to each region standardized by ISO 2631-1.



**Figure 12** Transmissibility response to unpaved road test

The frequency-domain transmissibility results shown in Figure 12 provide clear insight into the vibration attenuation efficacy of FAD relative to passive. The axle-to-chassis, axle-to-seat,

axle-to-lumbar, and axle-to-cervix transmissibility plots show prominent low frequency transmission peaks reduction when the FAD is employed.



**Figure 13** IMF energy response from DEP-CEEMDAN-MPE analysis

The CEEMDAN analysis provides an additional resolution of these frequency dependent changes. The Figure 13 reveals the improved performance of FAD configuration compare to passive. The reduction is particularly evident in the higher energy low frequency components of the chassis and lumbar responses. The inset in the chassis result further highlights the clustered low frequency IMF components, where the FAD exhibits lower integrated squared acceleration than the passive configuration. The cervix results showed dominated performance of FAD throughout the ride spectrum The CEEMDAN analysis supports the PSD observations by demonstrating that the reduction in vibration is distributed across the ride spectrum.

Overall, the unpaved road tests results demonstrate consistent attenuation across the principal evaluation layers. The PSD indicates reduced frequency-domain vibration content, the transmissibility results show reduced transmission from the axle through the vehicle and occupant-related locations, and the CEEMDAN analysis identifies lower integrated squared acceleration in the corresponding decomposed low-frequency components. These frequency-domain observations are accompanied by reductions in both frequency-weighted RMS acceleration and VDV, indicate that the vibration attenuation produced by the FAD is clearly reflected throughout the test.

## 5. Conclusion

Vehicle vibration transmitted to occupants is governed by the magnitude, frequency, duration, and direction of the excitation, and its effect can become particularly important for vulnerable occupants such as patients transported in medical emergency vehicles. The suspension and damping system therefore play a critical role in limiting the transmission of road vibration to the vehicle body and therefore occupants. These requirements become increasingly demanding under irregular and transient road excitation, motivating the development of damping technologies capable of providing improved vibration attenuation over a broad range of operating conditions.

This study experimentally evaluated the vibration attenuation performance of a frequency-adaptive damper (FAD) in a large van under a discrete bump and an unpaved-road condition. The FAD was compared with a conventional passive damping configuration using frequency-weighted RMS acceleration, vibration dose value (VDV), vibration transmissibility, power spectral density (PSD), and DEP–CEEMDAN–MPE-based IMF energy analysis. The vehicle chassis, seat, lumbar and cervix locations were selected for the study. The lumbar and cervix regions were selected as key anatomical locations for evaluating human-body vibration response, as vibration transmitted to these regions provides an important indication of the biodynamic response of the seated occupant.

Under bump excitation, the FAD demonstrated its most pronounced attenuation at the cervix measurement location, where the frequency-weighted RMS acceleration, VDV, and axle-to-cervix transmissibility were reduced by 34.0%, 54.3%, and 33.5%, respectively. Under unpaved-road excitation, the FAD maintained consistent attenuation under broadband and non-stationary vibration, with the most notable reduction in VDV observed at the cervix location (20.2%). PSD and DEP–CEEMDAN–MPE analysis further confirmed that the attenuation was concentrated within important low-frequency vibration components.

These results demonstrate that the FAD can improve ride quality by reducing the magnitude and transmission of road induced vibration to the vehicle and occupant related locations. The presented results provide a basis for further investigation of the potential benefits of FAD technology for occupant health and long duration WBV exposure. Future studies will investigate the performance of the FAD under extended exposure durations and a broader range of operating conditions, with particular emphasis on established health-related WBV measures and the biomechanical responses of the lumbar and cervix regions.

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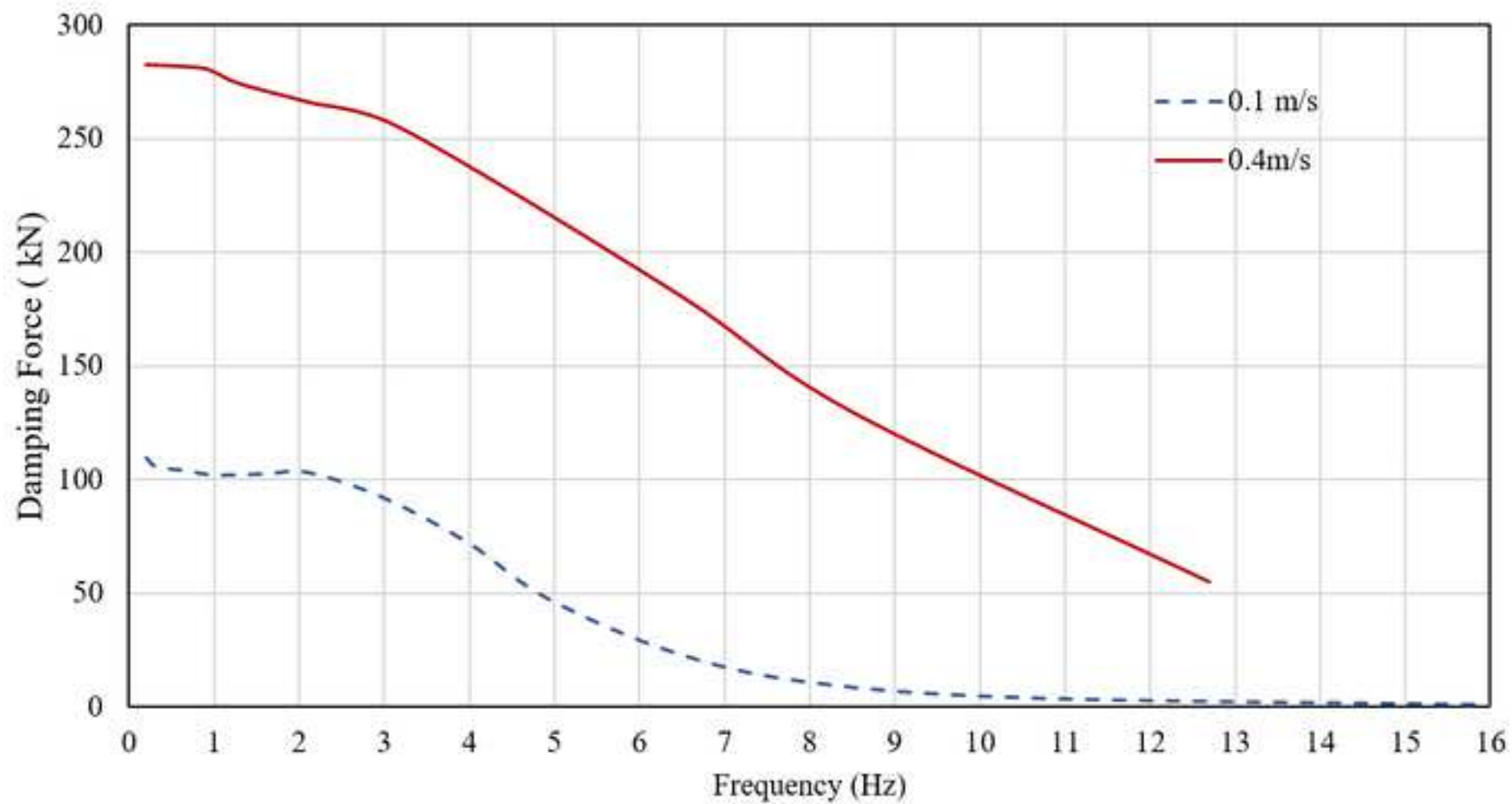


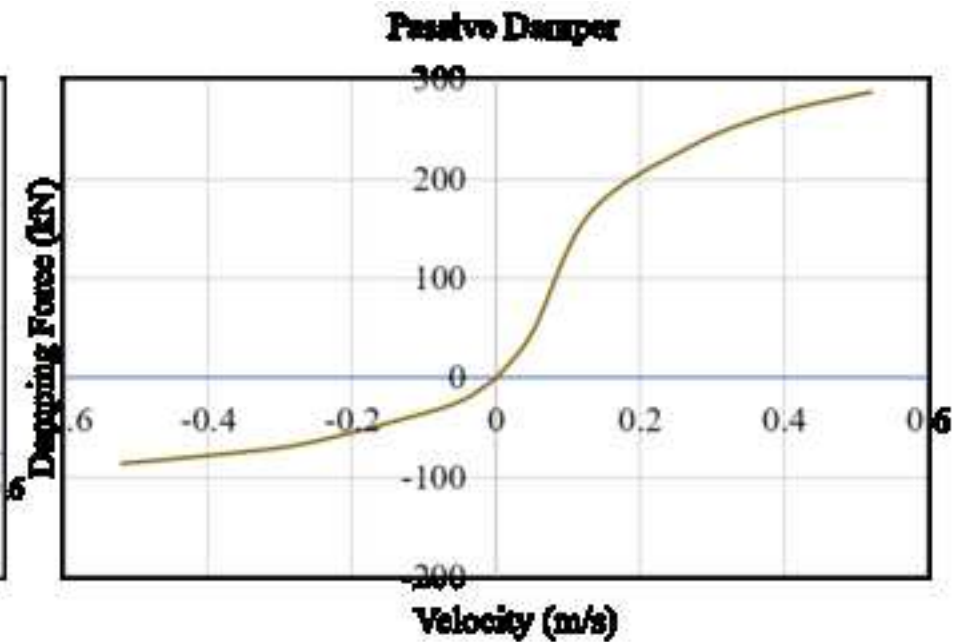
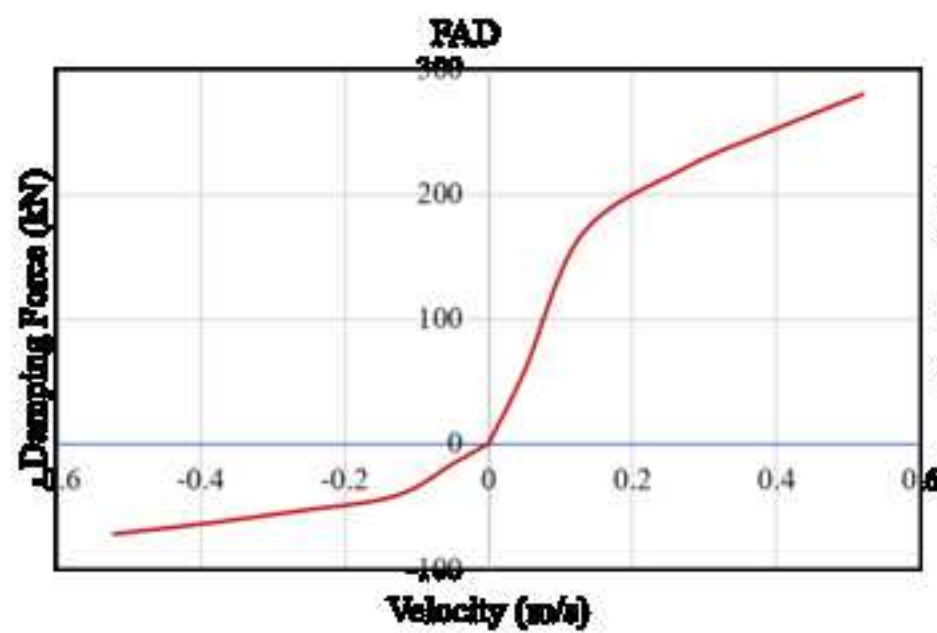


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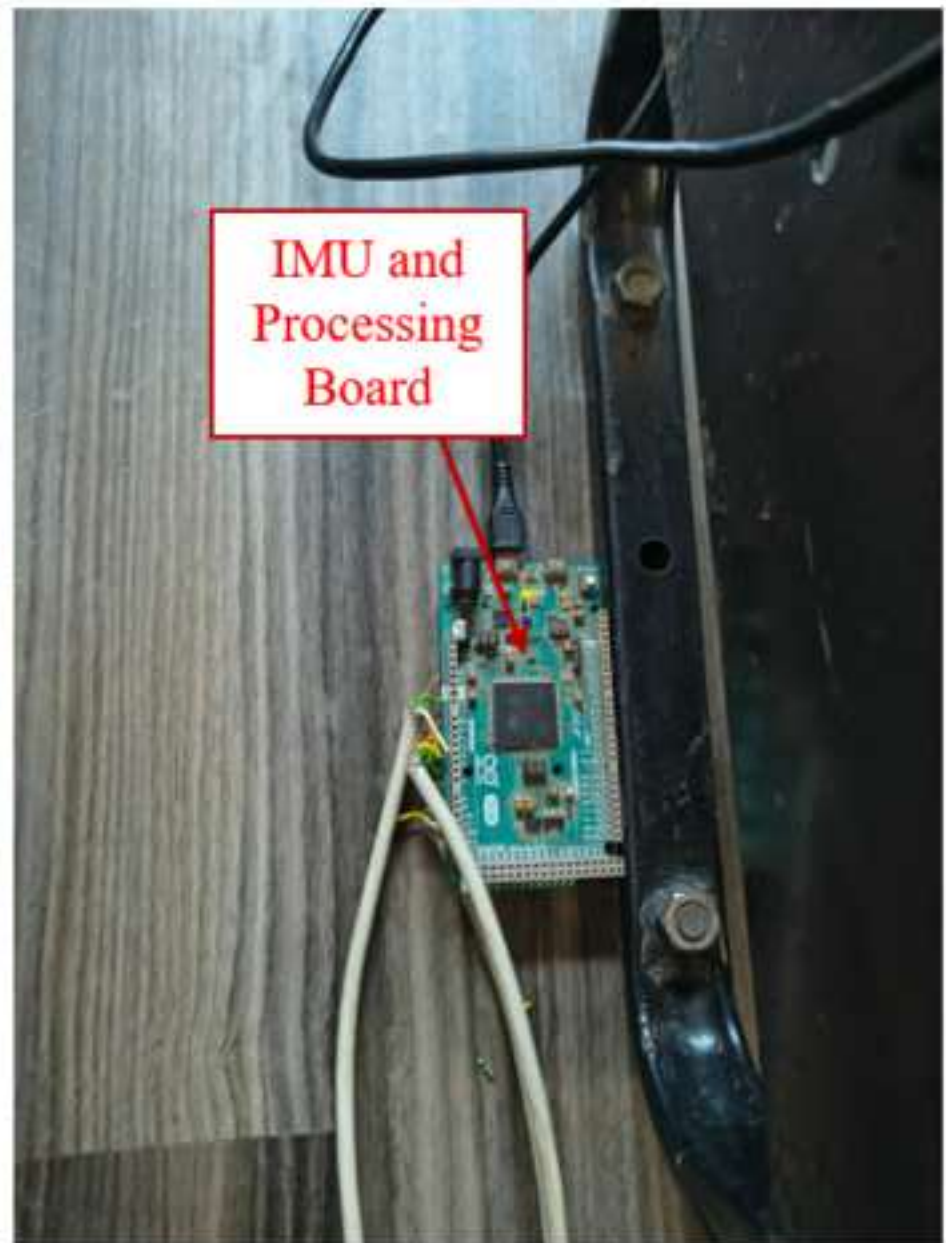
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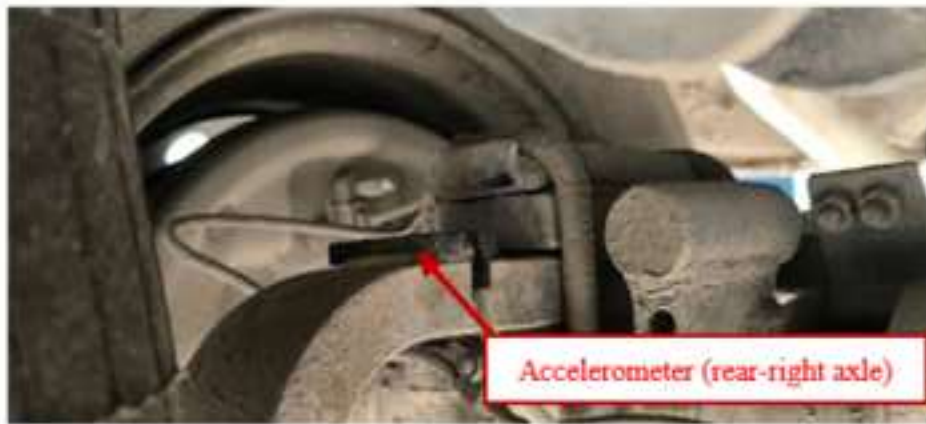




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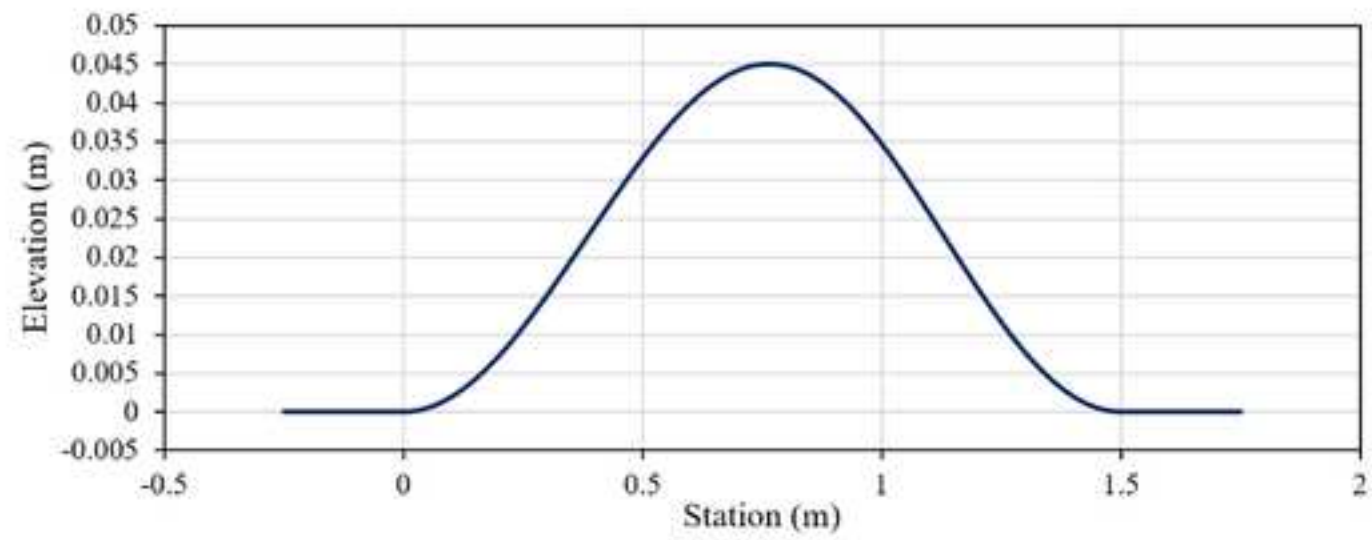
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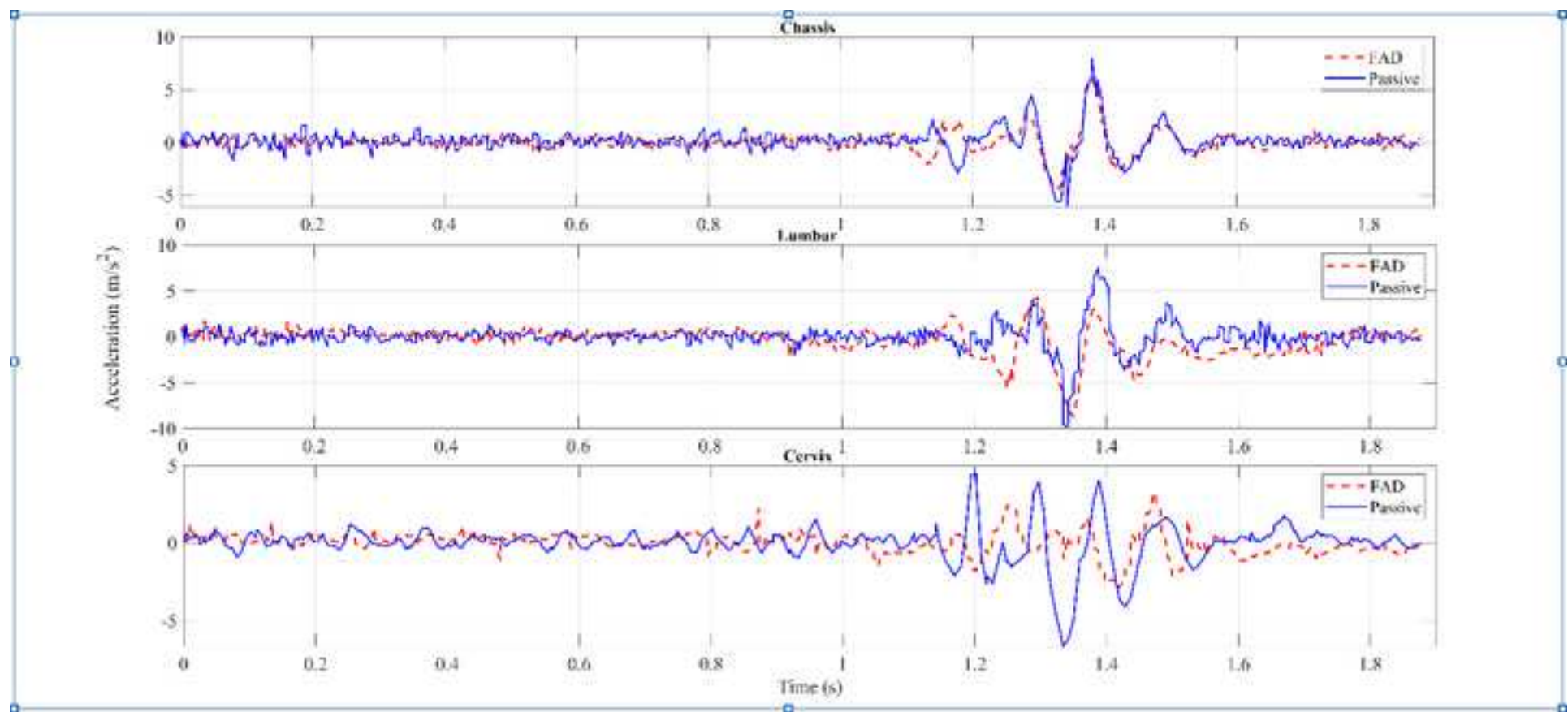


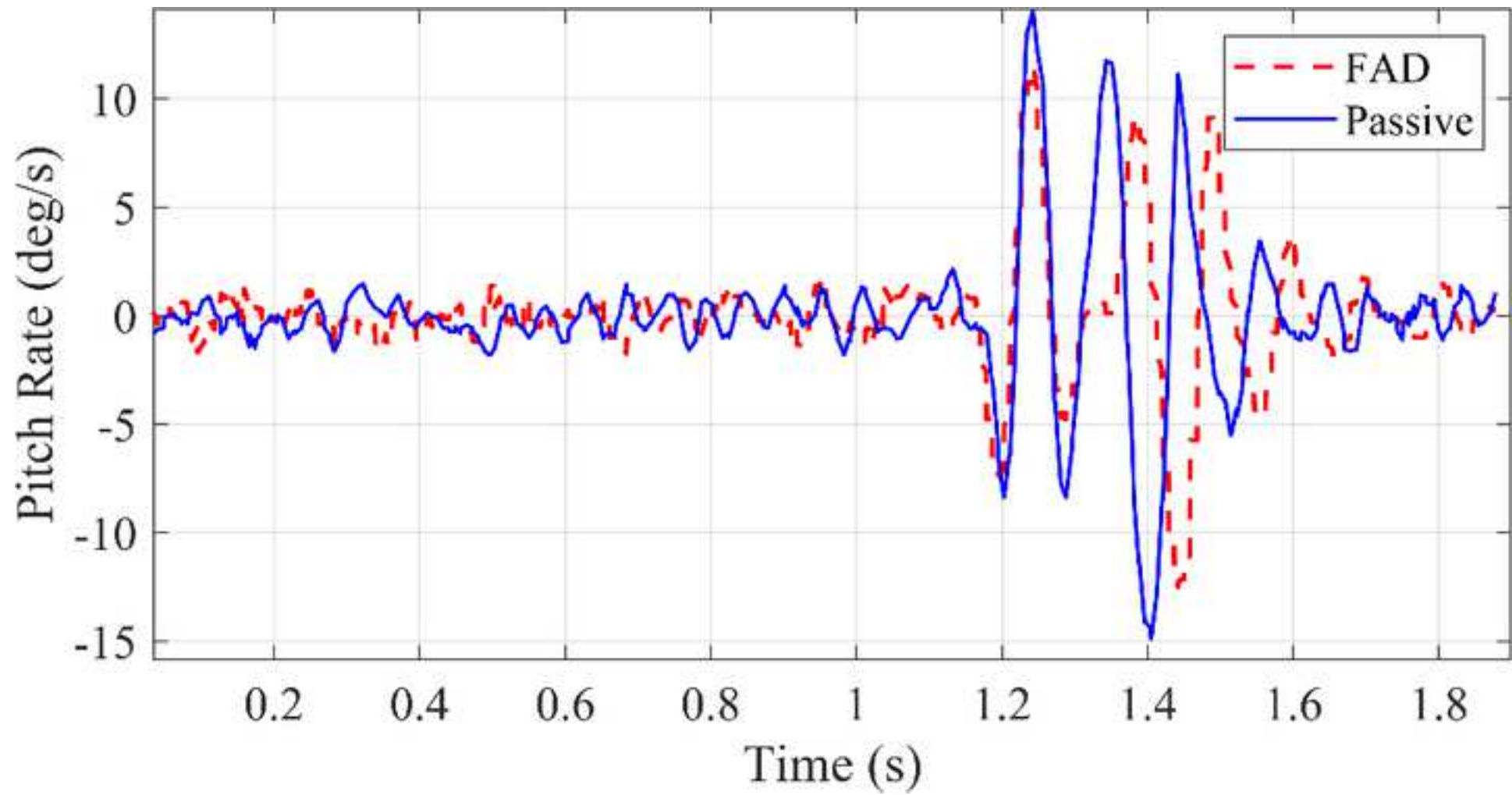
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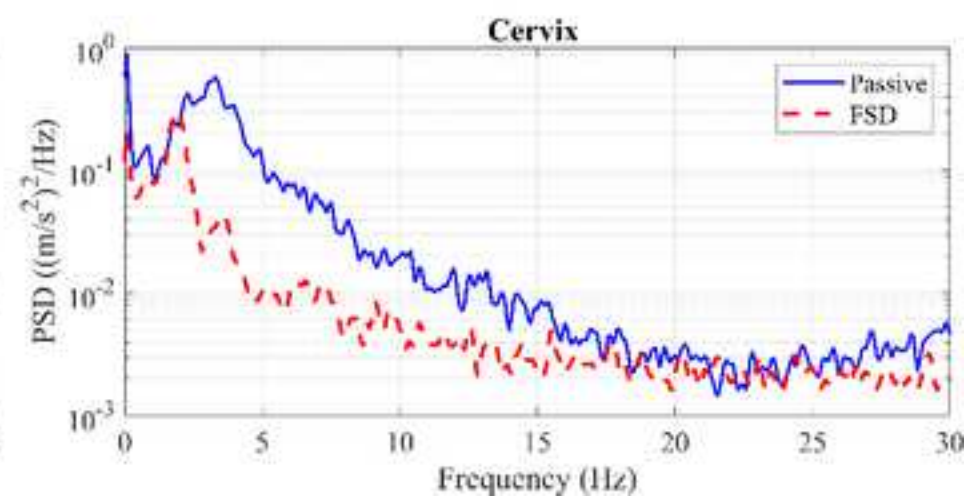
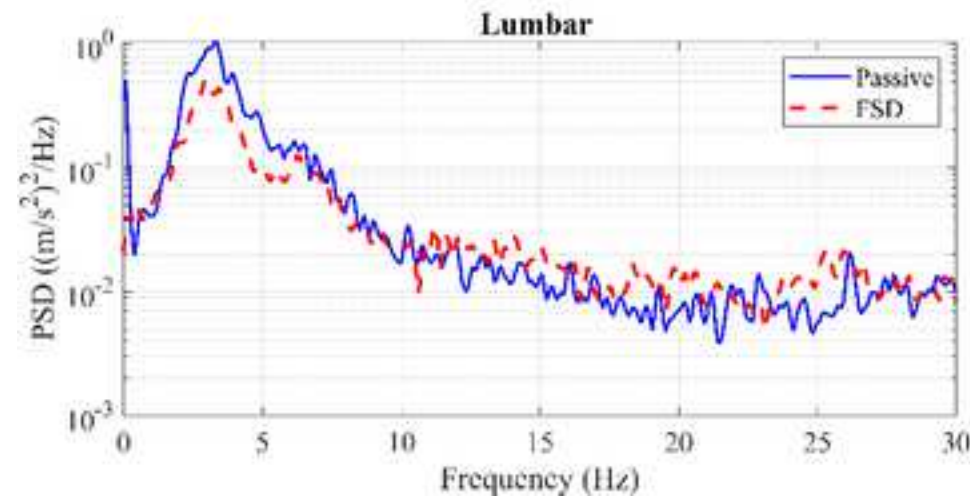
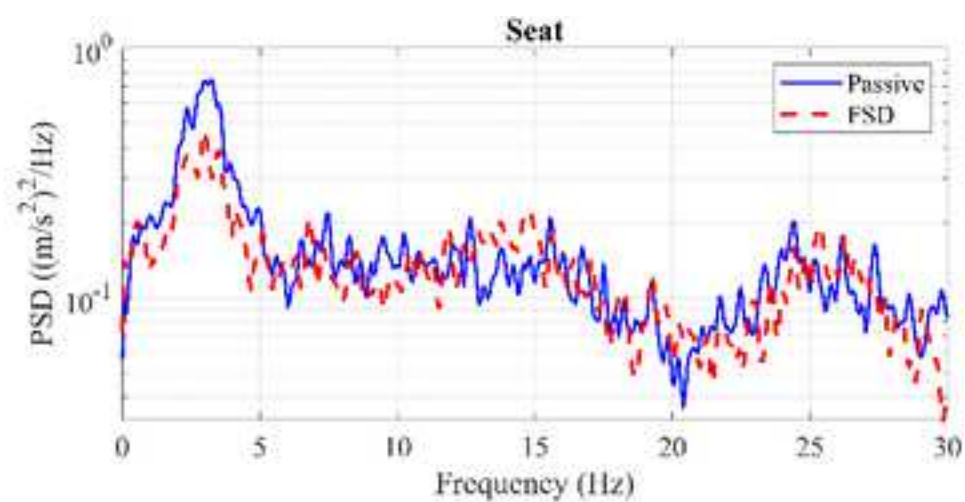
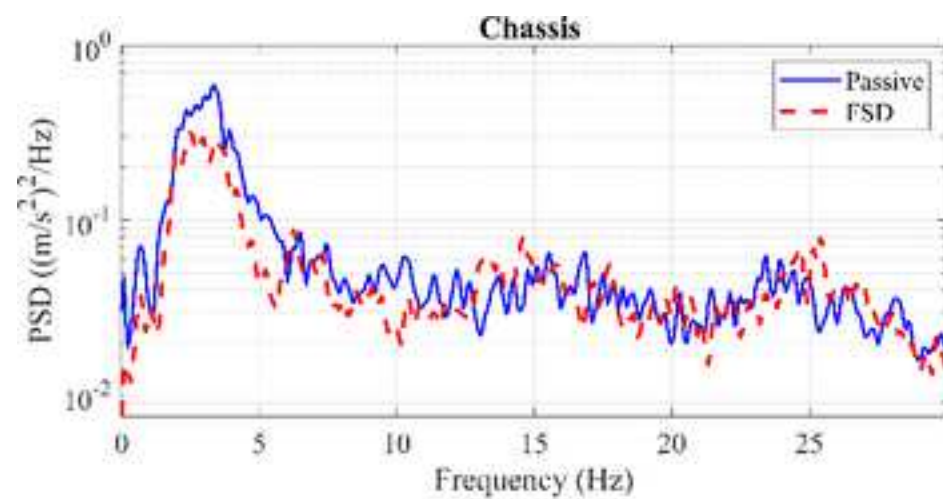


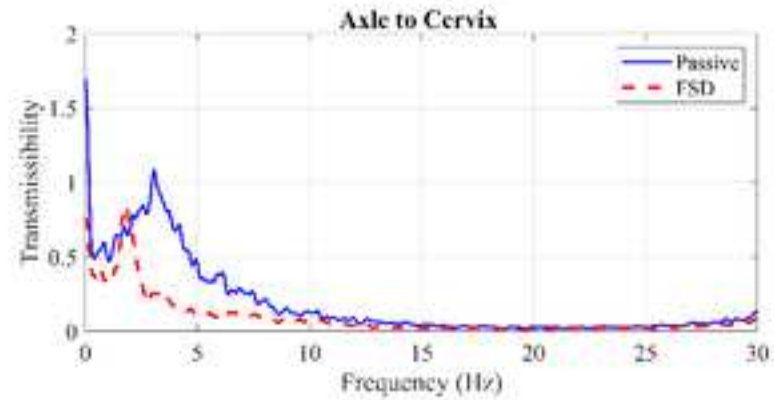
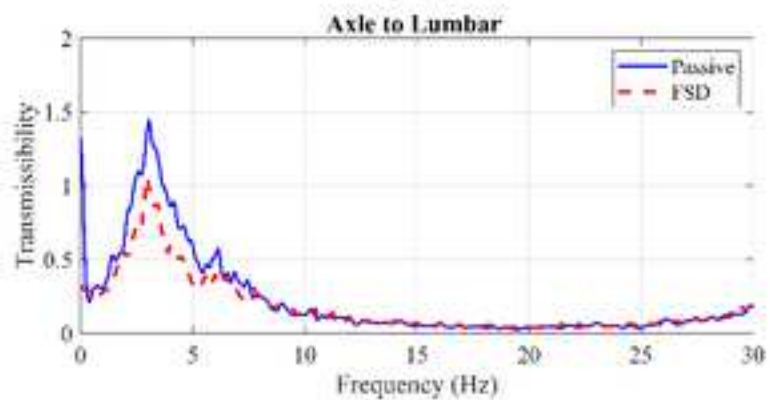
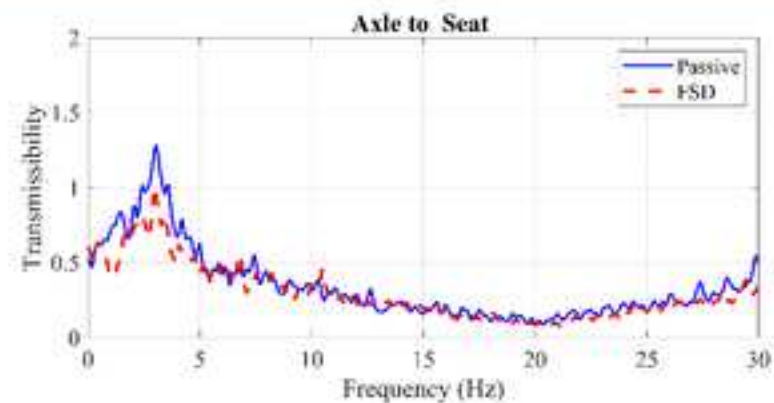
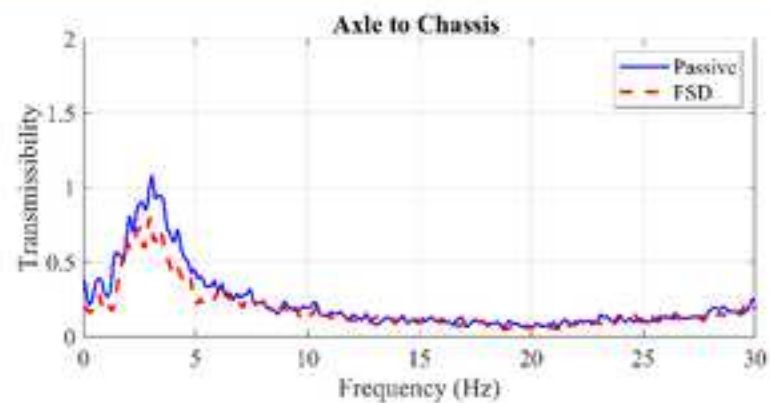
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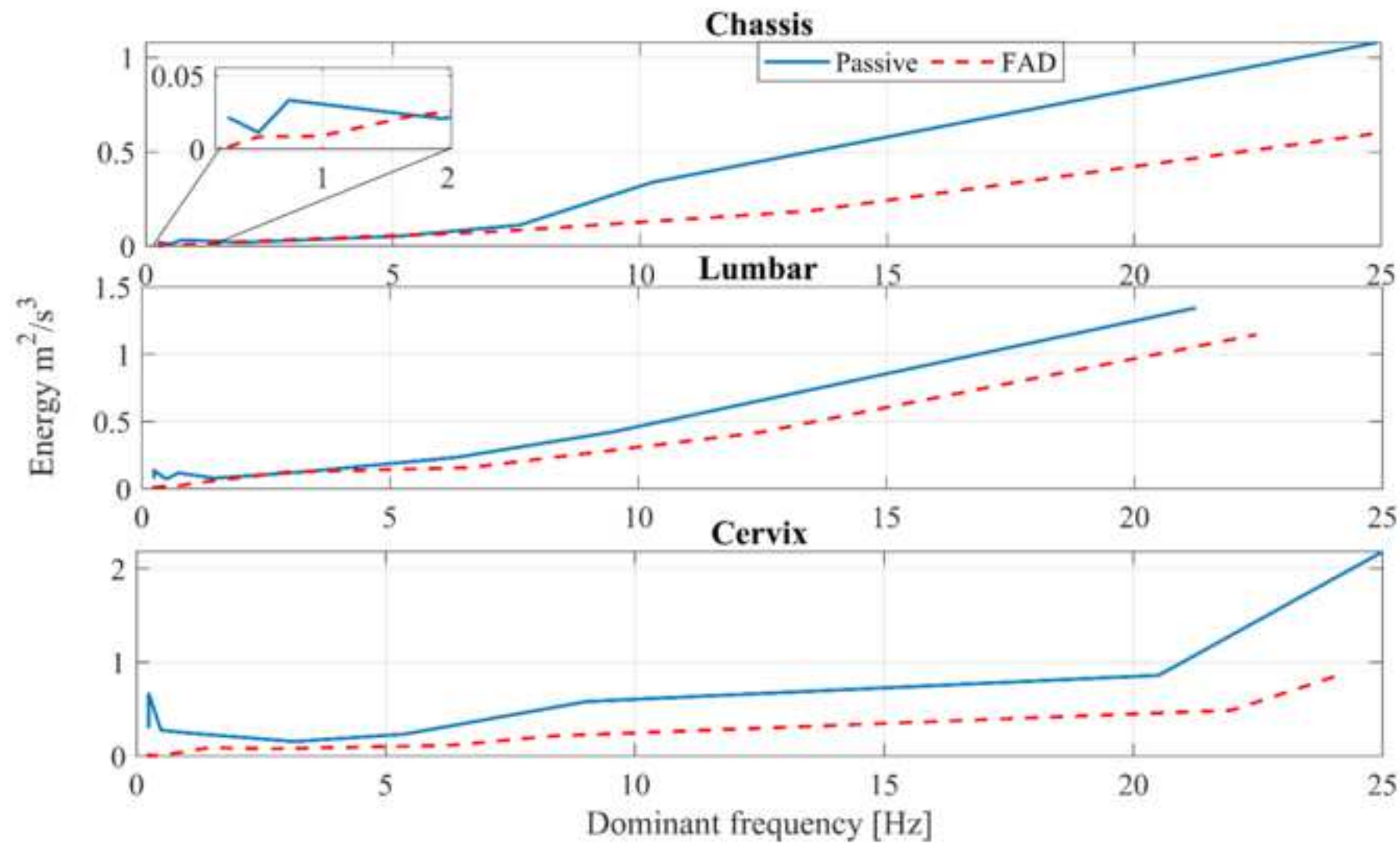












**Table 1** Test vehicle suspension parameters

| Parameter                   | Value                  |
|-----------------------------|------------------------|
| Sprung mass                 | 3080 kg                |
| Unsprung mass – front       | 186 kg                 |
| Unsprung mass – rear        | 244 kg                 |
| Spring rate – front         | 54,000 N/m             |
| Spring rate – rear          | 42,550 N/m             |
| Pitch inertia               | 4116 kg m <sup>2</sup> |
| Roll inertia                | 1029 kg m <sup>2</sup> |
| Front-axle distance from CG | 1.571 m                |
| Rear-axle distance from CG  | 1.779 m                |
| Track width – front         | 1.625 m                |
| Track width – rear          | 1.595 m                |

**Table 1** Specifications of the ADXL335 accelerometer and BNO055 IMU

| Sensor  | Type                        | Range               | Sensitivity / Resolution               | Sampling Frequency |
|---------|-----------------------------|---------------------|----------------------------------------|--------------------|
| ADXL335 | 3-axis analog accelerometer | ±3 g                | 300 mV/g                               | 1000 Hz            |
| BNO055  | IMU                         | ±4 g /<br>±2000 °/s | 14-bit accelerometer; 16-bit gyroscope | 100 Hz             |

**Table 1** ISO2631-1 based performance comparison of Passive and FAD during bump road test

|         | wRMS    |       |             | VDV     |       |             |
|---------|---------|-------|-------------|---------|-------|-------------|
|         | Passive | FAD   | % Reduction | Passive | FAD   | % Reduction |
| Chassis | 0.624   | 0.518 | 17.0%       | 3.349   | 2.562 | 23.5%       |
| Lumbar  | 1.105   | 0.985 | 10.9%       | 3.956   | 3.537 | 10.6%       |
| Cervix  | 0.656   | 0.433 | 34.0%       | 3.17    | 1.45  | 54.3%       |

**Table 1** ISO2631-1 based performance comparison of Passive and FAD during unpaved road test

|         | wRMS    |       |             | VDV     |       |             |
|---------|---------|-------|-------------|---------|-------|-------------|
|         | Passive | FAD   | % Reduction | Passive | FAD   | % Reduction |
| Chassis | 1.23    | 1.03  | 16.3%       | 4.58    | 3.771 | 17.7%       |
| Lumbar  | 1.425   | 1.187 | 16.7%       | 5.42    | 4.60  | 15.1%       |
| Cervix  | 1.159   | 0.994 | 14.2%       | 4.245   | 3.386 | 20.2%       |